

$T =$ operator on f.dim v.sp V over field F $\dim V = n$ λ -espace of T

v evector of $T \iff \begin{cases} v \neq 0 \text{ and} \\ Tv = \lambda v \text{ for some } \lambda \in F \end{cases} \iff (\lambda I - T)v = 0 \iff v \in \ker(\lambda I - T)$

λ evalue of T (for some λ -evector v) $\iff \ker(\lambda I - T) \neq 0 \iff p_T(\lambda) = 0$ where

$p_T(\lambda) = \det(\lambda I - T)$ regarded as a polynomial in variable λ , called char. poly of T

$$p_T(\lambda) = \lambda^n - \text{Trace}(T)\lambda^{n-1} + \dots + (-1)^n \det(T)$$

T has at most n distinct evals.
 If it does, T is automatically diagonal. (Converse false)

Independent of basis

$$\begin{aligned}
 & \det(\lambda I - \tilde{P}^{-1}AP) \\
 &= \det(\tilde{P}^{-1}(\lambda I)P - \tilde{P}^{-1}AP) \\
 &= \det(\tilde{P}^{-1}(\lambda I - A)P) \\
 &= \det(\tilde{P}^{-1}) \det(\lambda I - A) \det(P)
 \end{aligned}$$

T is diagonalizable
 (ie can be represented
 by a diagonal matrix)

$\iff V$ has an eigenbasis
 corresponding to T

$$V = \bigoplus_{\lambda \in F} \ker(\lambda I - T)$$

$$\iff \dim V = \sum_{\lambda \in F} \dim \ker(\lambda I - T)$$

Thm Let $Tv_i = \lambda_i v_i$ $i=1, \dots, k$
 λ_i distinct. Then

$$\sum v_i = 0 \Rightarrow \text{each } v_i = 0$$

("Distinct eigenspaces of T are linearly independent")

Pf. Induction on k . $k=1$ ✓

let ① $v_1 + \dots + v_k = 0$ Apply T

$$\text{② } \lambda_1 v_1 + \dots + \lambda_k v_k = 0$$

$$\text{②} - \lambda_1 \text{① gives } (\lambda_2 - \lambda_1)v_2 + \dots + (\lambda_k - \lambda_1)v_k = 0$$

By induction $(\lambda_j - \lambda_1)v_j = 0 \quad \forall j=2, \dots, k$

As $\lambda_j - \lambda_1 \neq 0$ we get $v_j = 0$ —//—

$\therefore v_1 = 0$ too.

(Rmk. Can apply $T - \lambda_1 I$ directly)

Operator T is called nilpotent if $T^N = 0$ for some $N > 0$.

Example:

$$T = \begin{bmatrix} 0 & 1 & & 0 \\ & 0 & 1 & \\ & & \ddots & \ddots \\ 0 & & & 0 \end{bmatrix}$$

$$0 \leftarrow e_1 \leftarrow e_2 \leftarrow \dots \leftarrow e_n$$

Observe $T^n = 0$ ($T^{n-1} \neq 0$)

Claims (A) T nilpotent and diagonalizable $\Rightarrow T=0$

(B) ① T is nilpotent $\iff$ ② T has an upper/lower Δ matrix and diag entries are 0
 $\iff$ ③ $P_T(x) = x^n \iff$ ④ $T^{\dim V} = 0$

Pf (A) Note for diag. operator T , $P_T(x) = (x-\lambda_1) \cdots (x-\lambda_n)$ where λ_i are the eigenvalues (possibly repeats)

For nilpotent T , let v be an evector then $Tv = \lambda v \Rightarrow T^k v = \lambda^k v \Rightarrow T^N v = 0 = \lambda^N v \Rightarrow \lambda^N = 0 \Rightarrow \lambda = 0$

So all evals of T are 0. The only way to be diagonalizable is $T=0$

(B) ① $\xleftrightarrow{\checkmark}$ ④ $\xleftrightarrow{\checkmark}$ ② $\xleftrightarrow{\checkmark}$ ③

easy parts

Pf of ② $\Rightarrow$ ④ { If T has strictly upper Δ matrix w/ basis $v_1, v_2, \dots, v_n$

$$\left\{ \begin{array}{l} Tv_1 = 0 \\ Tv_2 = a_{12}v_1 \\ Tv_3 = a_{13}v_1 + a_{23}v_2 \\ \dots \end{array} \right\} \Rightarrow T^i v_i = 0 \quad \forall i$$

Quiz next Mon! (D) For any operator T on V , we
portion details: have the Fitting decomposition
on Thu $V = \ker(T^\infty) \oplus \text{Im}(T^\infty)$

(C) $p_T(x)$ factors fully
 $\Updownarrow$
Has a triangular matrix
(This gives (3) $\Rightarrow$ (2))

Consider Idea for proof of (1) $\Rightarrow$ (2) Use one of the chains below and extend basis!

$$V \supset TV \supset T^2(V) \supset \dots \supset T^{N-1}(V) \supset T^N(V) = 0 \quad \text{When } T \text{ is nilpotent}$$

or consider

$$0 \subset \ker(T) \subset \ker(T^2) \subset \dots \subset \ker(T^{N-1}) \subset \ker(T^N) = V$$

$\uparrow$
 $\#$
 V

Tasks for tomorrow
Try!
• Prove C
• Complete B
• Prove D